INSTITUTE OF SPACE TECHNOLOGY ISLAMABAD



Electrical Network Analysis Lab

EE-19B

PROJECT TITLE: *VISUAL AC MAINS VOLTAGE INDICATOR*

SUBMITTED TO:

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Objective of Our Project:

- To vary the base voltage of transistors using variable resistors.
- To convert the supplied AC voltage to DC by help of a half wave rectifier (one diode) and a filter (one capacitor).
- DC and AC has direct relation when we vary AC so DC also varies and this change is sensed by transistors and further indicated by LEDs.
- We are visualizing the change in AC voltages by potentiometers, transistors, and LEDs.

Components Description and Use in our Project:

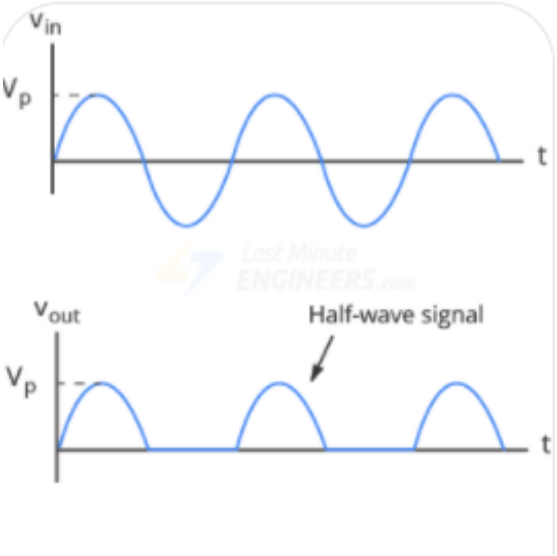
- **Diode (D1):**

➢ **Explanation of element:**

A diode is a device that allows current to flow in one direction but not the other. This is achieved through a built-in electric field.

➢ **Use in our circuit:**

In our circuit it is used as a half wave rectifier as we see that the ac voltage supplied from secondary winding of transformer is converted to pulsating DC as diode becomes reverse biased during -ve AC half cycle, omitting the -ve part of AC wave.



- **Capacitor (C1):**

➢ **Explanation of element:**

A capacitor (originally known as a condenser) is a passive two-terminal electrical component used to store energy electrostatically in an electric field. Unlike a resistor, a capacitor does not dissipate energy. It stores a certain level of supplied DC voltage and function as a temporary energy storage device. It allows AC current and block DC current.

- **Use in our circuit:**

  In our circuit the capacitor is used as a filter because it makes our rectified half wave closer to dc by removing ripples in the wave. Here the capacitor converts the pulsating DC to pure DC.

- **Resistors (R1 to R6):**

- **Explanation of element:**

  A resistor is a passive two terminal electrical component that implements electrical resistance. It is used to reduce current flow, adjust signal levels, divide voltages, bias active elements, and terminate transmission lines among other uses.

- **Use in our circuit:**

  Here 3 resistors are connected in series with 3 LEDs so that it doesn't draw too much current from the source and burn out. To absorb the difference between the supply voltage and the rated voltage of the LED and to protect it from voltage fluctuations.
  However here we are placing a 1k ohm resistor in series to a potentiometer so that even if the potentiometer is set at 0 ohms then the series resistor prevent too much current from flowing through it.

- **Zener Diodes:**

- **Explanation of element:**

A Zener diode is a special type of diode designed to reliably allow current to flow backwards when a certain set reverse voltage known as Zener voltage is reached.

- **Use in our circuit:**
  Here the Zener diodes are used as voltage regulators (keep voltage constant) they maintain the dc Zener voltage between $I_{Z(min)}$ and $I_{Z(max)}$.

- **Transistor (T1 to T3):**

- **Explanation of element:**

Transistor is a semiconductor device used for amplifying, controlling, and generating electrical analysis. The small current that we apply at the base makes a big current to flow between the collector and emitter. This is a one direction device, flows current in one direction.

- **Use in our circuit:**
  Because of transistors the LEDs glow because the transistors sense the change in input AC voltage and that change is expressed in the form of glowing of the LEDs.

When we give a certain AC input voltage then by adjusting the transistor respective potentiometer we provide the required voltage at the transistor base hence the transistor starts conducting and LED glows the LED is turned off when voltage at base is dropped.

- **Potentiometer:**

  - **Explanation of element:**

A potentiometer is a 3 terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.

  - **Use in our circuit:**

In our circuit we use our variable resistor to supply the required voltage at the base of transistor for LED to glow. Without it we cannot adjust transistor base current so LEDs won't glow.

- **Transformer:**

  - **Explanation of element:**

    A transformer is an electrical apparatus designed to convert alternating current from one voltage to another. It can be designed to step up or step down voltages and works on the magnetic induction principle.

  - **Use in our circuit:**

    We have a 9 V – 9 V transformer and this transformer is used to step down our AC voltage present at the primary winding of the transformer. And that step down AC voltage is provided at the secondary winding.

- **230V AC voltage supply:**

  - **Explanation of element:**

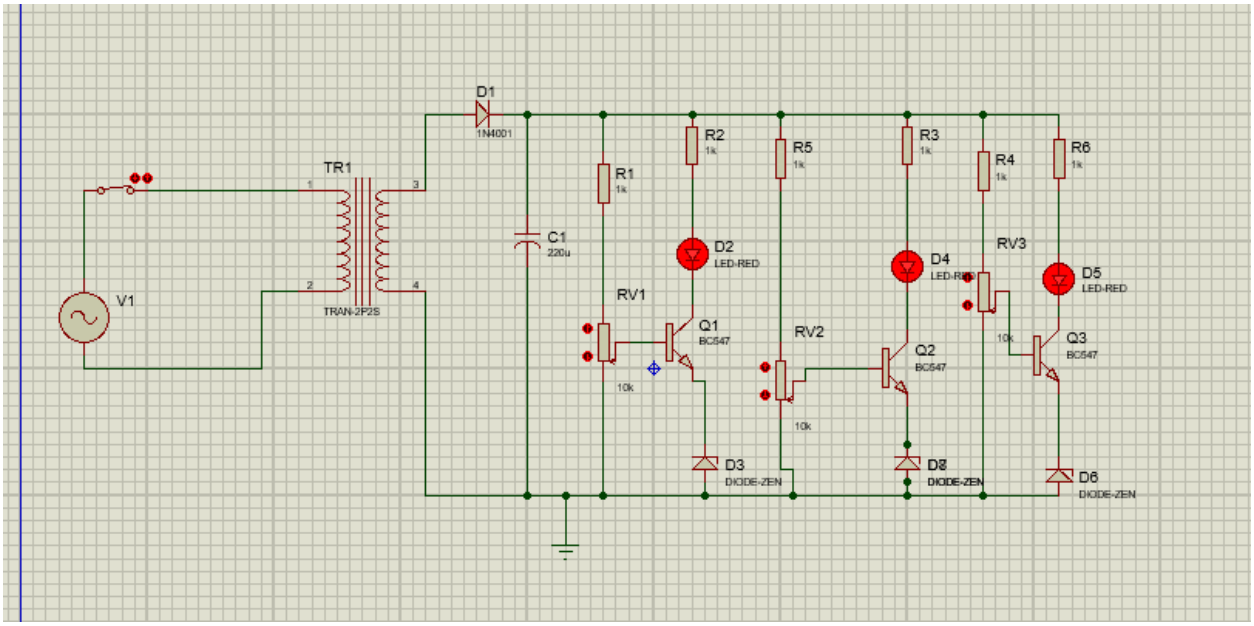
    230 V AC is a voltage between 220 to 240 V with a frequency of 50 Hz, this voltage is provided from power sockets and plugs.

  - **Use in our circuit:**

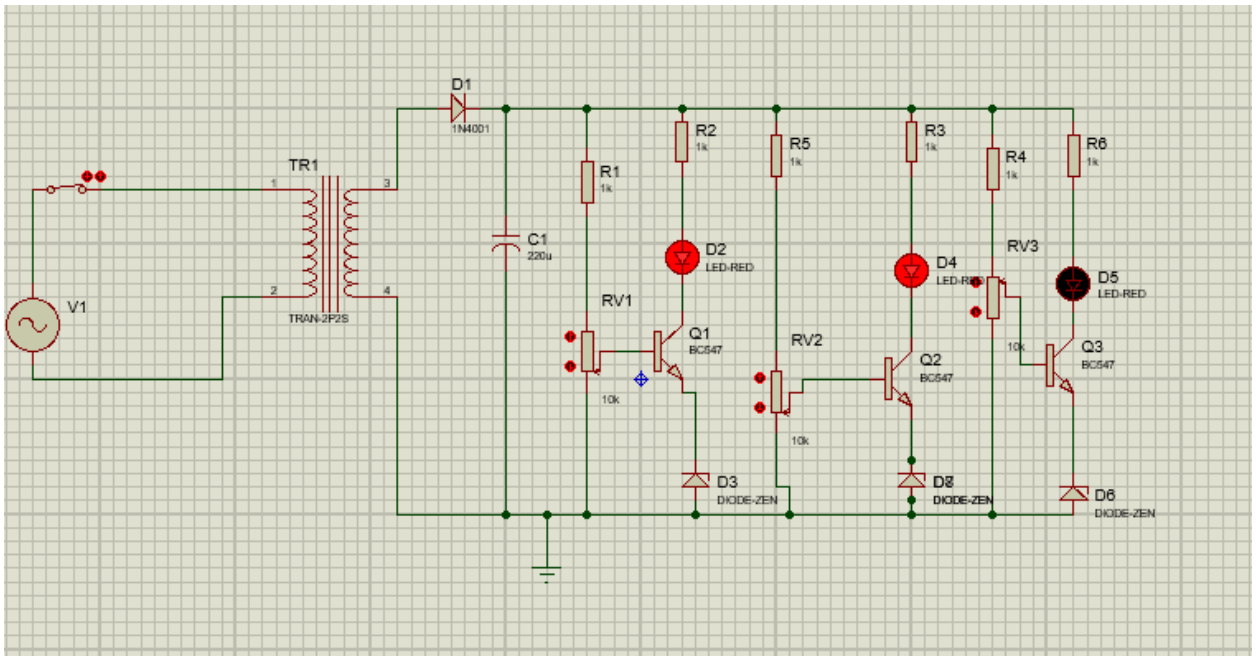
    We aur providing supply voltage from the 230 V AC and next to it we had placed a switch when its open connection of supply to circuit will break if its connected so supply will remain.

- **Circuit Schematic:**

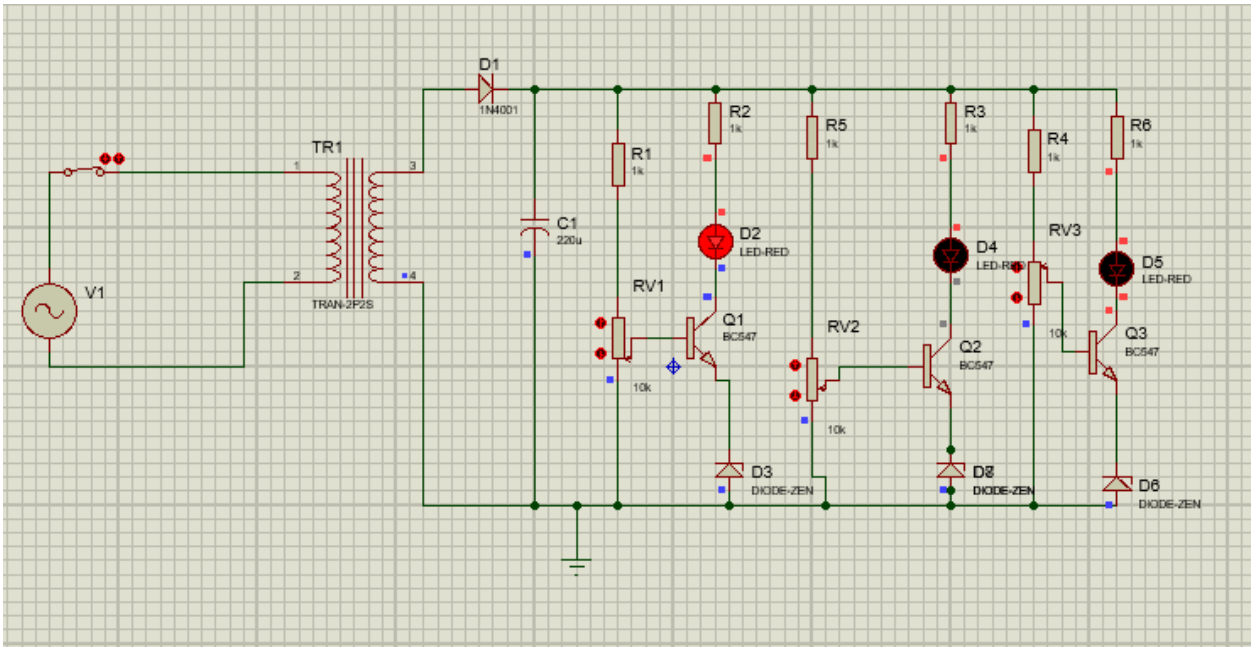
  - **When input is greater than 240V:**



➢ **When input is greater than 200 and 230V:**



➢ **When input is greater than 175 and 200V:**



- **Conclusion of this project:**
  - When voltage drop is below 175 volts no led will glow
  - When the voltage drop is between 200V to 230V this voltage range is shown by two LEDs.
  - When voltage drop is between 175-200V .It is indicated by glowing of 1 led.
- **Applications of this project:**
  - A simple AC main voltage indicator circuit gives visual indication of AC mains voltage from 160V to 270V in steps of 10V. For input mains voltage less than 160V all LEDs are off.
  - It's ideal application (advanced circuit) is in housing area of middle or low class society and company buildings which are using high electricity consumption, but at the same time cannot afford to use a voltage stabilizer. LEDs turning on and off as voltage level transition from one to another is dynamic without perceptible delay.